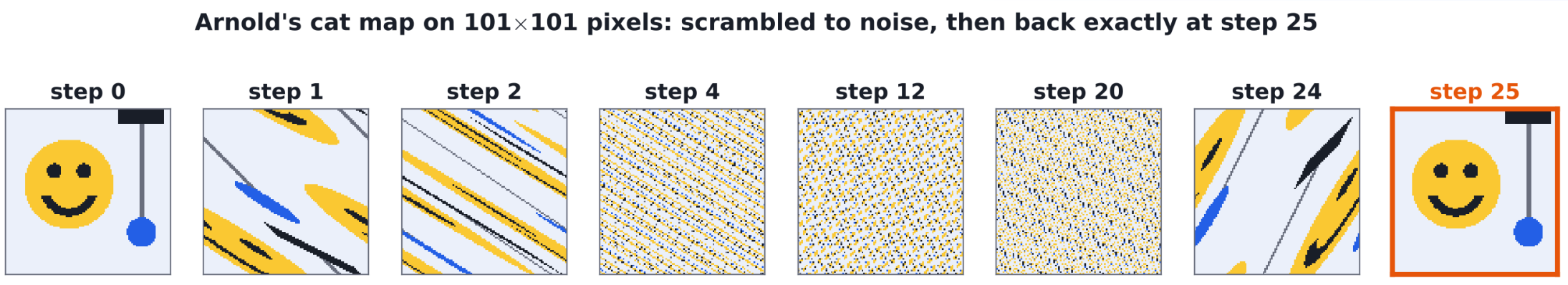
Everything Comes Back (Eventually)

The Poincaré Recurrence Theorem



[Name 1] · [Name 2] · [Name 3] · [Name 4] | Classical Mechanics course project, IIT Bombay

The theorem

Let T map a space Ω to itself, with **finite volume** $\mu(\Omega) < \infty$ and **no squeezing**: $\mu(T^{-1}A) = \mu(A)$ for every set A.

Then for any set A with $\mu(A) > 0$, **almost every** point of A comes back to A, and does so **infinitely often**.

In plain words: wait long enough and almost every state returns as close as we like to where it started.

- **Finite room**: the system cannot run off to infinity
- **No squeezing**: volume of a set of states stays fixed
- **Almost every**: the exceptions have zero volume

Poincaré found this in 1890 while working on the three-body problem.

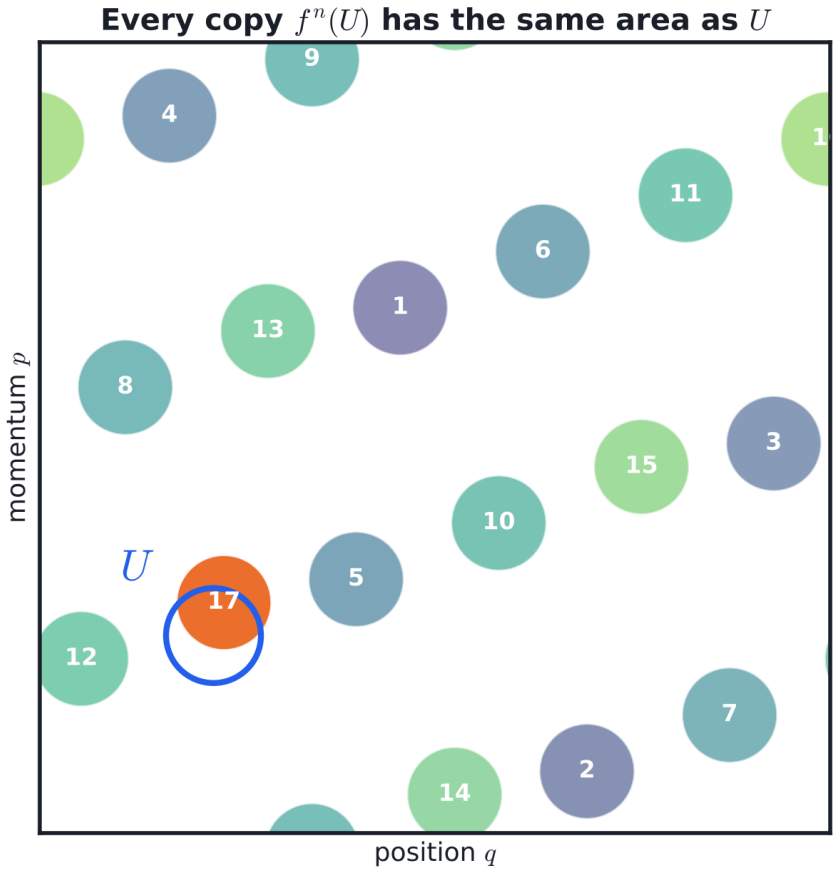
Proof in one picture: the pigeonhole idea

Let $B \subset A$ be the points that **never** return to A.

- The sets $B,\ T^{-1}B,\ T^{-2}B, \ldots$ never overlap. If two did, a point of B would come back to B.
- Each has the same volume $\mu(B)$.
- They all fit inside Ω, so

$$\sum_{k=0}^{\infty} \mu(B) \ \leq \ \mu(\Omega) < \infty \ \Rightarrow \ \mu(B) = 0.$$

Infinitely often: the points that return only finitely many times also form a zero-volume set, by the same trick.



Copy 17 lands on U. Finite room plus equal-sized guests means sharing.

Why physics cares: Liouville's theorem

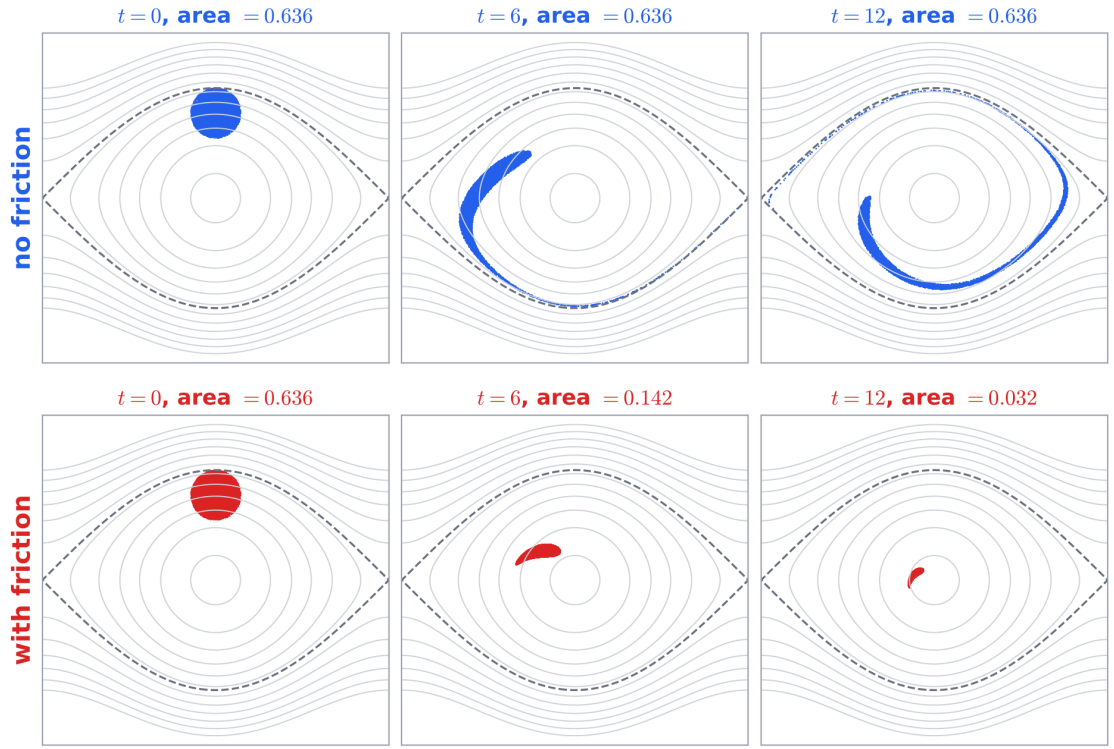
Hamilton's equations: $\dot{q} = \partial H / \partial p$, $\dot{p} = -\partial H / \partial q$

The flow in phase space has zero divergence:

$$\frac{\partial \dot{q}}{\partial q} + \frac{\partial \dot{p}}{\partial p} = \frac{\partial^2 H}{\partial q \, \partial p} - \frac{\partial^2 H}{\partial p \, \partial q} = 0$$

So **volume is kept**. Fixed energy in a box keeps the **room finite**. Both conditions hold.

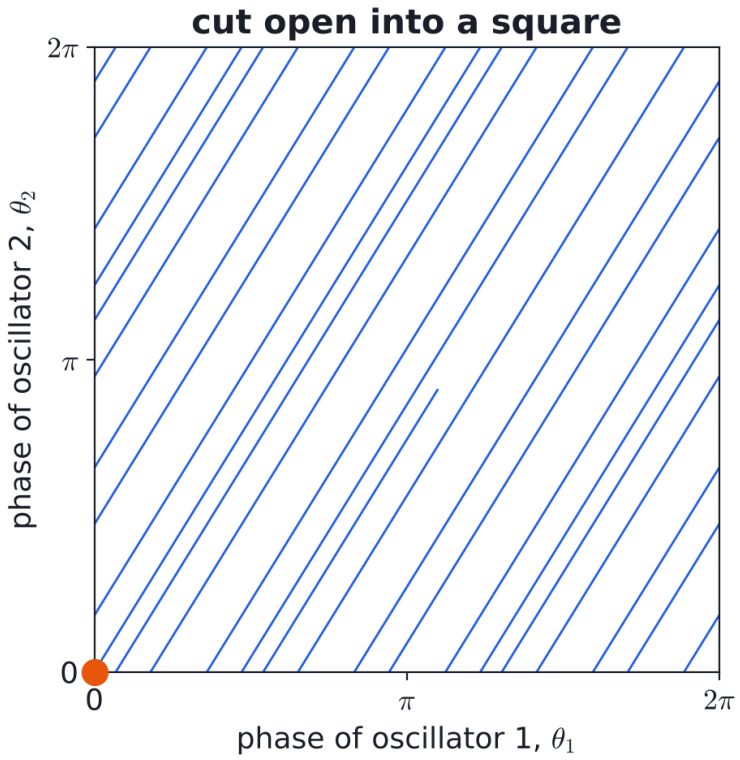
> Add friction $-\gamma p$: divergence becomes $-\gamma$, area shrinks like $e^{-\gamma t}$. No recurrence.



A blob of pendulums. Area without friction: 0.63617 at the start, 0.63616 at the end.

Example 1: two oscillators with an irrational ratio





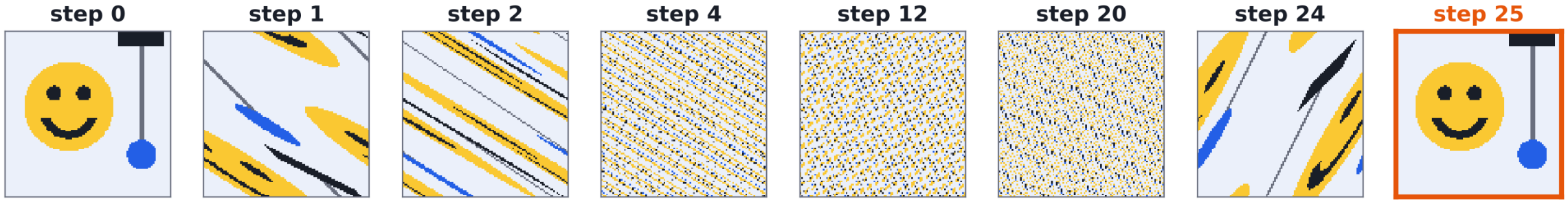
- Phases $\theta_1 = \omega_1 t$, $\theta_2 = \omega_2 t$, ratio $\omega_2/\omega_1 = \varphi = 1.618\ldots$
- The state lives on a **torus**. The path **never closes**, yet comes back as close as we like

- Mean wait to return within ε: $T \propto \varepsilon^{-1}$ for 2 oscillators, ε^{-2} for 3
- Simulation matches $T = (2\pi)^n / (|\omega|\, \sigma_\varepsilon)$ to within 3%

Example 2: chaos does not stop it (Arnold's cat map)

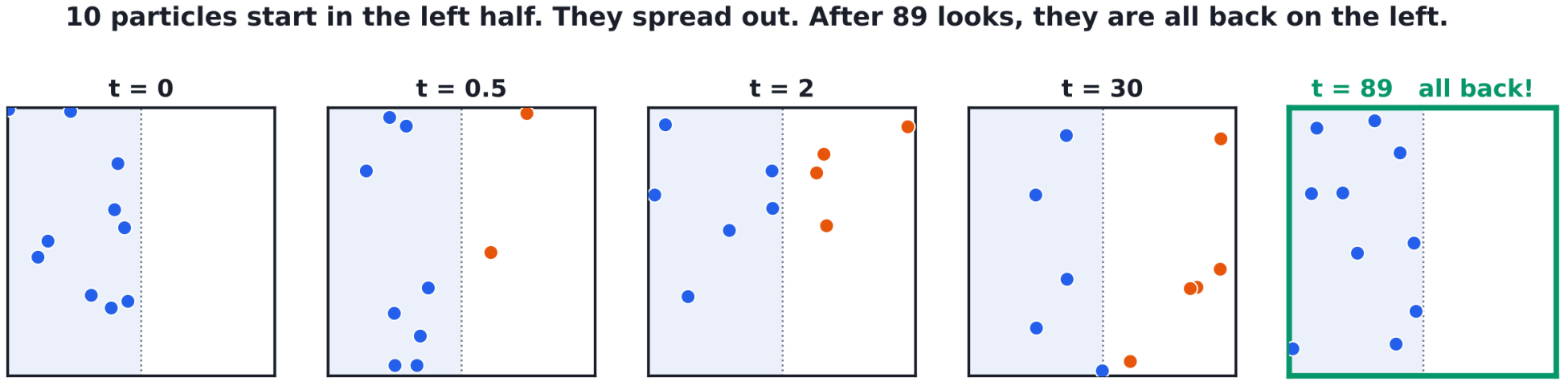
$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \bmod N, \qquad \det = 1 \;\Rightarrow\; \text{no squeezing}$$

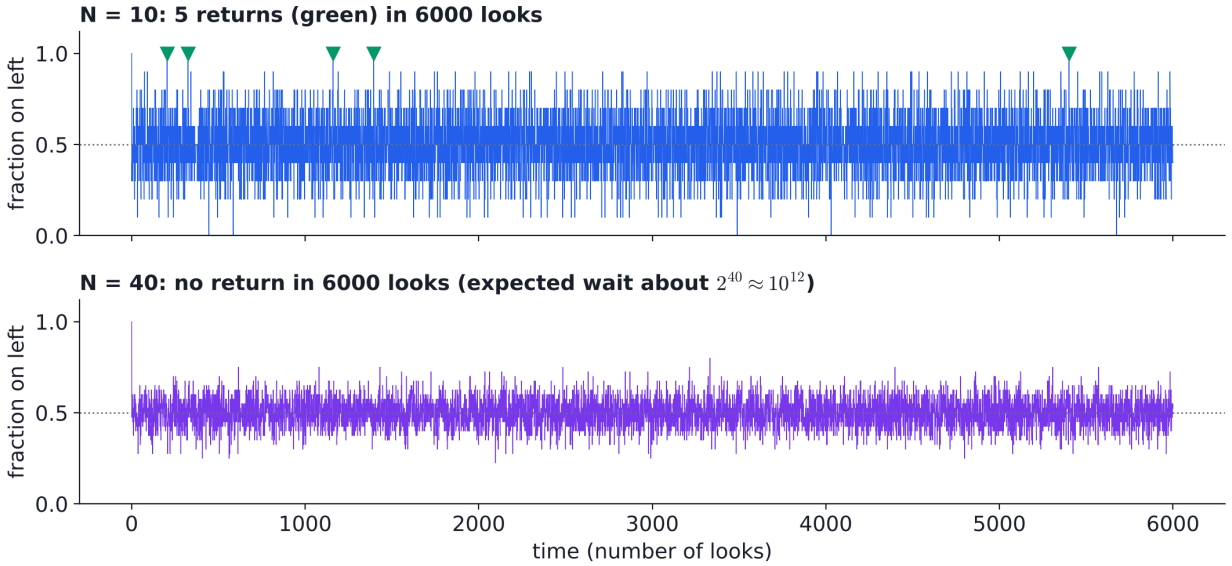
Arnold's cat map on 101×101 pixels: scrambled to noise, then back exactly at step 25



- Very chaotic: lengths stretch by $\frac{3+\sqrt{5}}{2} \approx 2.6$ every step
- Still, a shuffle of a **finite** set of pixels must repeat
- On a 101×101 grid the face returns **exactly** at step 25
- Cats always come home. This one just takes the scenic route

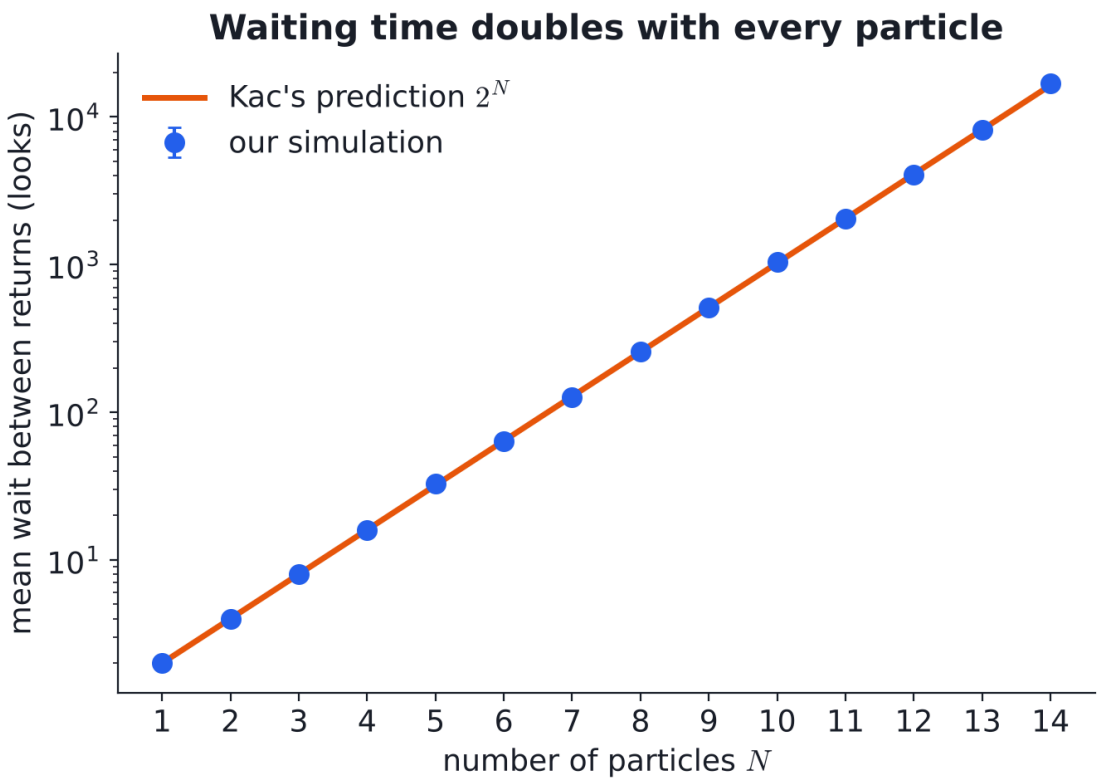
Our numerical case: a gas in a box





- N free particles bounce off the walls. All start on the **left**
- Unfold each bounce: the gas becomes a point on an N-torus. Finite, no squeezing, so **they must come back**
- $N = 10$: back 5 times in 6000 looks.
  $N = 40$: not once

How long is "eventually"?



"All on the left" takes up $\mu(A) = 2^{-N}$ of the torus. **Kac's lemma**:

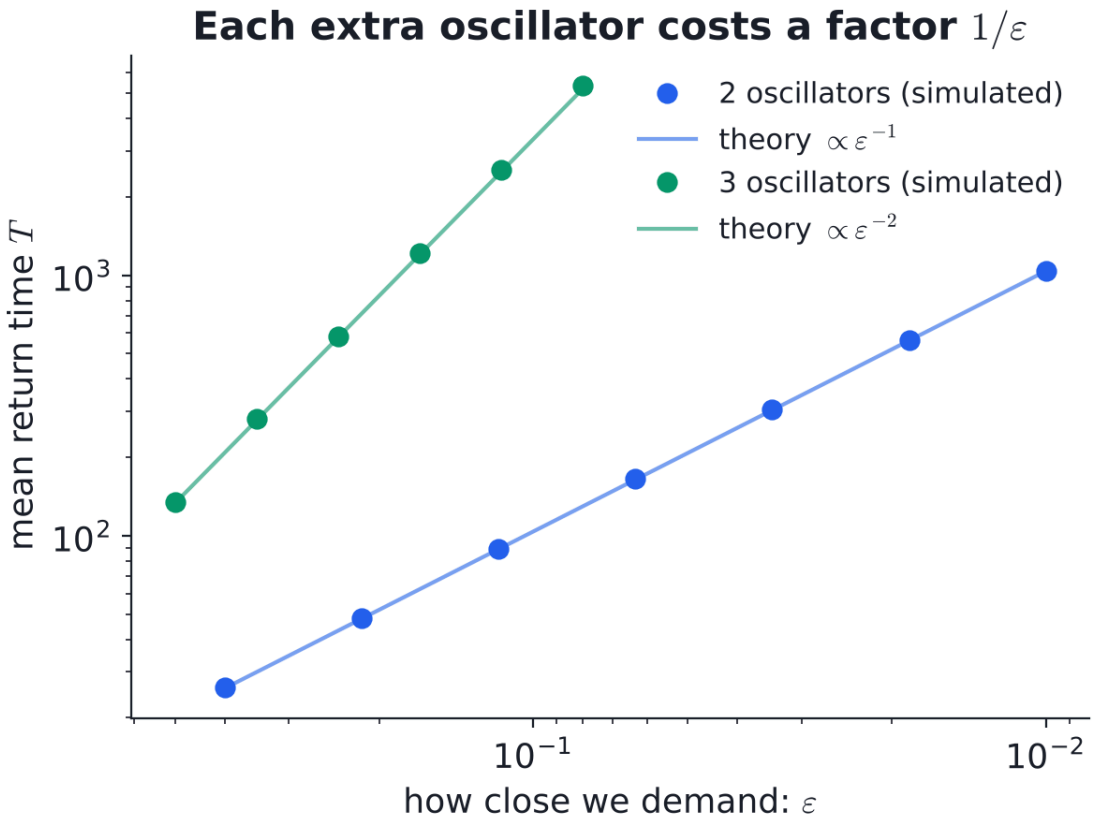
$$\langle T \rangle = \frac{1}{\mu(A)} = 2^N \text{ looks}$$

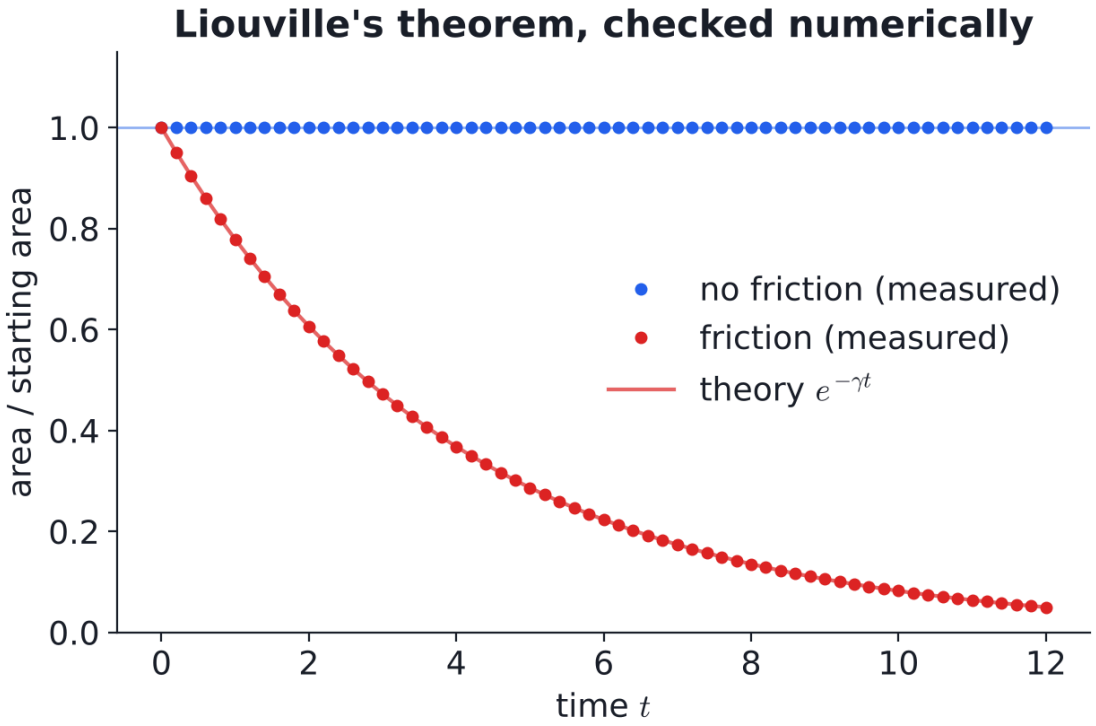
Our runs (about 4000 returns each) agree for $N = 1$ to 14.

$N = 100$, one look per millisecond: $\sim 4 \times 10^{19}$ years, about **3 billion times the age of the universe**. A room has $\sim 10^{27}$ molecules.

Recurrence is guaranteed, but for big systems it is slower than slow. That is how it lives in peace with the second law.

Backup: return time vs closeness





Left: mean return time to an ε-ball for 2 and 3 oscillators, with the flux formula. Right: blob area with and without friction against $e^{-\gamma t}$.

References: H. Poincaré, **Acta Math.** 13 (1890) · M. Kac, **Bull. AMS** 53 (1947) · V. I. Arnold & A. Avez, **Ergodic Problems of Classical Mechanics** (1968) · F. J. Dyson & H. Falk, **Amer. Math. Monthly** 99 (1992)